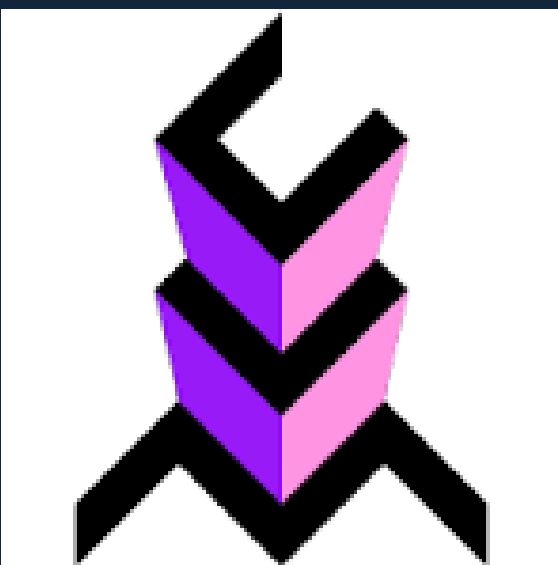




# WHY LARGE LANGUAGE MODELS FAIL FOR HAUSA EDUCATIONAL CONTENT: CASCADING ERRORS FROM TRANSLATION TO SPEECH TO COMPREHENSION

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## Motivation

Large language models (LLMs) fail catastrophically when applied to Hausa educational content due to cascading errors from machine translation (MT) to speech synthesis to automatic speech recognition (ASR) to question answering. We evaluate multiple models on HausaPQ-Speech, a new dataset of 4,703 WASSCE exam questions. MT exhibits 6.9–13.9% omission rates. ASR achieves 68–92% word error rate (WER) with 100% error on tonal vowels. LLM cascading for error correction yields minimal improvement (<15% WER) or causes hallucination. Question answering accuracy remains below 30%, with Hausa-specialized models underperforming general multilingual models. We conclude that fully automated systems are not viable for Hausa educational deployment; human-in-the-loop approaches remain essential.

| Stage                          | Output / Observation                                                                                                                                                                                                                                                                   |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| English source                 | "Which of the following is NOT classified as an animal? A. Amoeba B. Paramecium C. Euglena D. Obelia"<br>✓ All 4 answer options present and semantically correct.                                                                                                                      |
| After MT (Google Translate)    | "Euglena" rendered as <b>"Eugokena"</b> (transliteration error); option marker partially dropped; constraint "NOT" at risk of omission.<br>■ Omission rate: 6.9–13.9% across corpus (Table 18). TTR drops 17.3% → 13.9%.                                                               |
| After TTS → ASR (WER 89.47%)   | "wani idan ba a rara biyu halittu masu zuwa, kamar da ba a amai ba aron cu, shi igbo ke na da obaliya"<br>All 4 answer options lost. Tonal vowels: <b>100% error rate</b> across all ASR models.<br>✗ Question now unintelligible. Correct answer irrecoverable from transcript alone. |
| After Llama-3.3-70B correction | Replaced content with hallucinated string: <b>"Kimiya Noma 2010"</b> (generic subject label — not error correction).<br>WER: 98.25%   CER Δ: −22.92% (degradation).<br>✗ Correction made output worse. Consistent with option-omission hallucination pattern (Table 8).                |
| QA result (N-ATLaS)            | No valid answer letter extractable via regex. Marked <b>invalid</b> .<br>✗ Correct answer (C. Euglena) never recovered across any pipeline stage.                                                                                                                                      |

Table 1. An example of the five failure points: how MT transliteration, ASR collapse, and LLM hallucination compound to produce an invalid QA result in the N-ATLaS pipeline.

## Methodology

HausaPQ-Speech: 4,703 WASSCE questions (2010–2024, 10 subjects) with English text, Hausa MT (Google Translate), VITS TTS audio (4–73s), and answers.

### Evaluations:

- MT: BLEU, TER, omission rate
- ASR (940 samples): whisper-large-hausa, Hausa-ASR (NCAIR1), wav2vec2-hausa, whisper-large-v3
- LLM cascading (20 samples (95% CI ±10%)): Llama-3.3-70B, Mixtral-8x7B, Gemini-2.0-Flash, Flan-T5-Base → WER, CER
- QA (490 MCQs): Llama-3.2-1B, Gemma-2-2B, N-ATLaS, HausaLlama → accuracy, completion
- Metrics: BLEU, TER, omission rate, WER, CER, tonal vowel error rate

## Experimental Setup



Figure 1: Evaluation pipeline Results

The following results reveal that human-in-the-loop systems with native speaker validation remain the only viable path for real-world deployment. Future work must prioritise tone-aware ASR/TTS, structurally-aware MT, and robust QA evaluation.

| Model        | Accuracy ↑ | Completion |
|--------------|------------|------------|
| N-ATLaS      | 29.5%      | 42.2%      |
| Llama-3.2-1B | 27.4%      | 97.6%      |
| Gemma-2-2B   | 25.6%      | 99.8%      |
| HausaLlama*  | 25.3%      | 100.0%     |

Table 2 : LLM Accuracy on Hausa WASSCE Questions (n = 490)

| Model         | Corrected WER | WER Δ   | CER Δ   |
|---------------|---------------|---------|---------|
| Llama-3.3-70B | 98.25%        | +14.49% | −22.92% |
| Mixtral-8×7B† | 112.74%       | 0.00%   | 0.00%   |
| Gemini 2.0†   | 112.74%       | 0.00%   | 0.00%   |
| Flan-T5-Base  | 110.65%       | +2.09%  | −28.98% |

Table 3: ASR Error Correction via LLM Cascading (n = 20, 95% CI ±10%; baseline WER = 112.74% for all models) Positive WER Δ = improvement; negative CER Δ = degradation. † No valid outputs (0/20). Llama-3.3-70B hallucinated content rather than correcting errors.

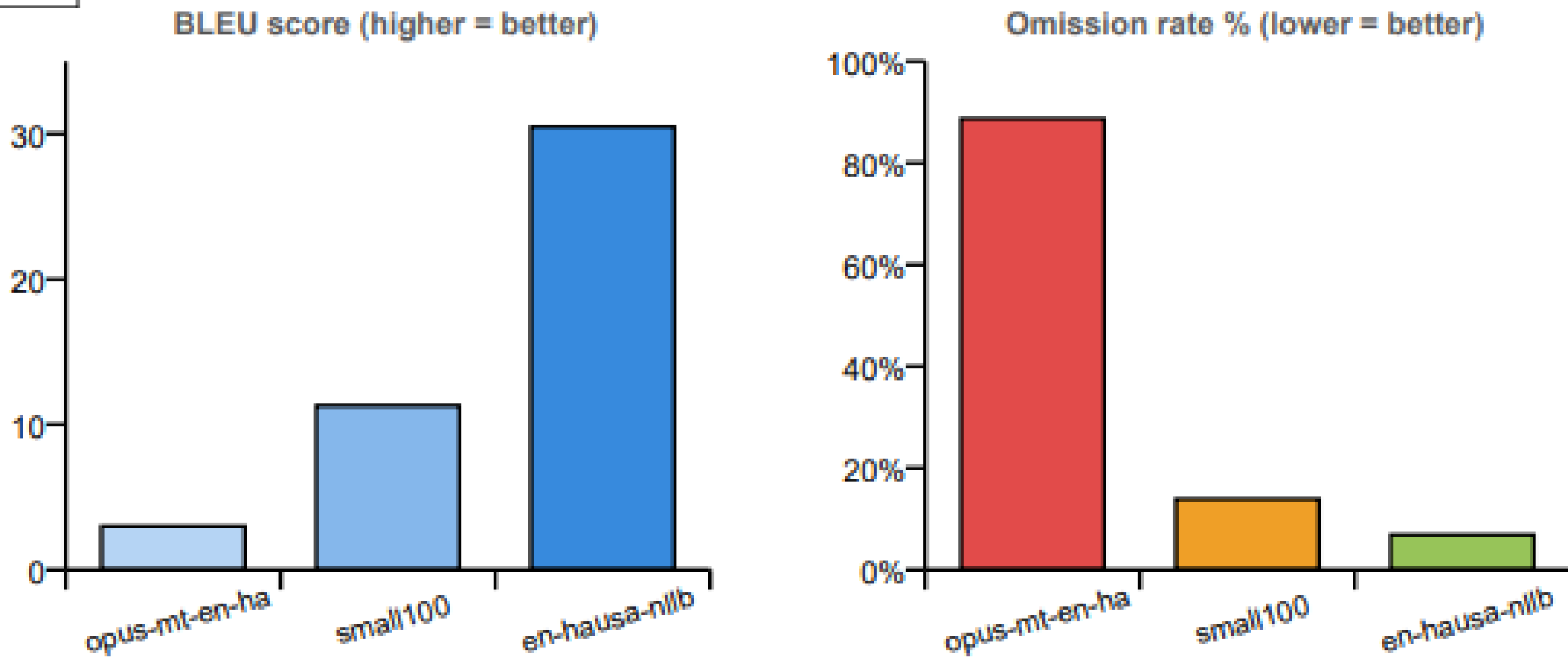


Figure 2: Comparison of three MT models on BLEU score (left) and omission rate % (right). Higher BLEU and lower omission rate indicate better performance.

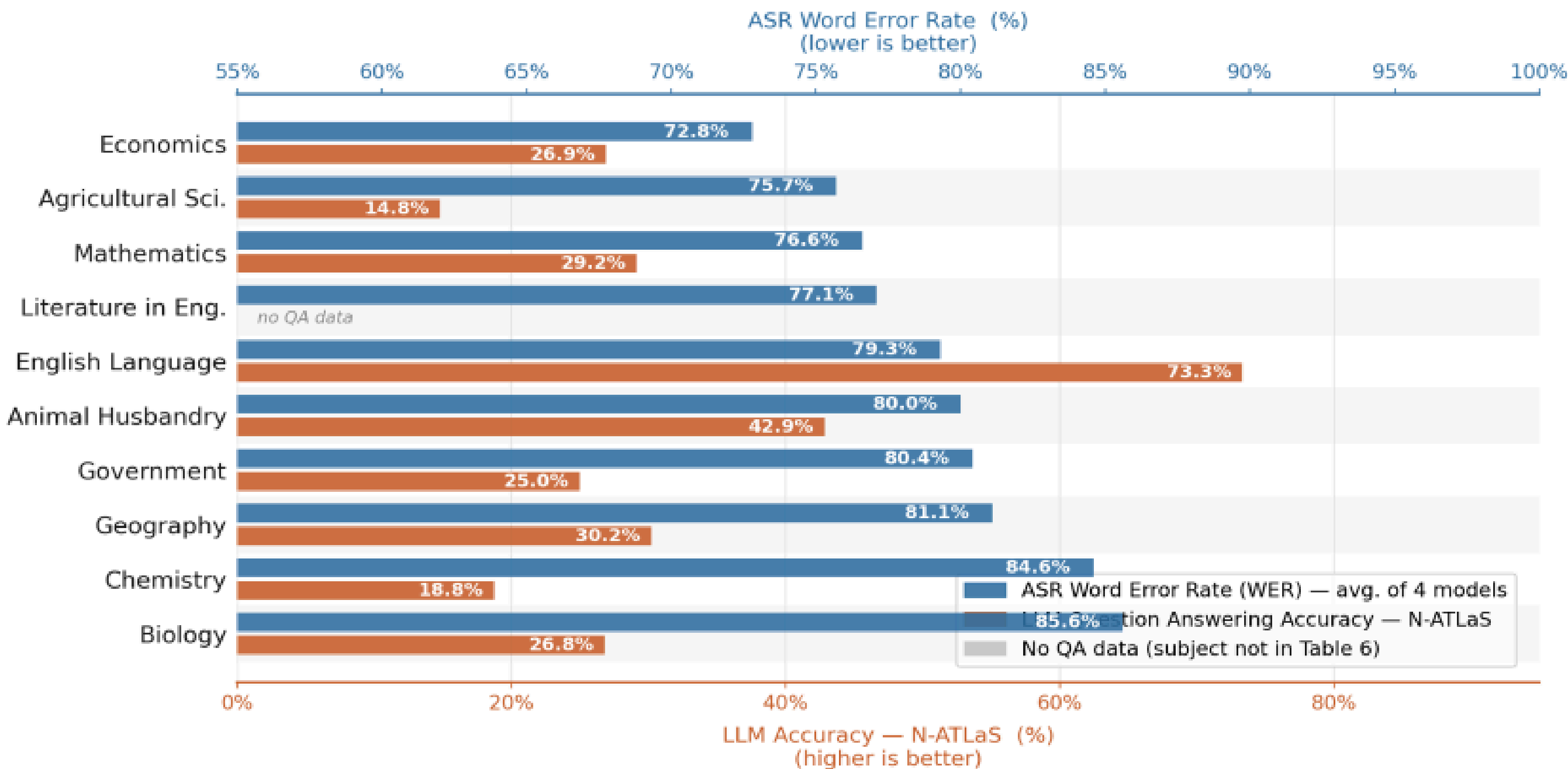


Figure 3: Per-subject ASR Word Error Rate (WER, cool colours, top axis) and LLM question-answering accuracy (warm colours, bottom axis) across 10 WASSCE subjects. WER averaged across four ASR models; accuracy from N-ATLaS on 490 Hausa multiple-choice questions.

## Conclusions

- MT omission rates: 6.9–13.9%, with lexical flattening (TTR dropped from 17.3% to 13.9%)
- ASR WER: 68–92%, 100% tonal vowel error rate across all models
- LLM cascading: Llama-3.3-70B gave +14.49% WER improvement but −22.92% CER degradation (hallucination); Mixtral and Gemini showed 0% improvement
- QA accuracy: 25–30%; N-ATLaS highest (29.47%) but 58% invalid completions; HausaLlama (Hausa-specialized) underperformed general models
- Simple rule-based baselines failed to improve WER; LLM cascading often made outputs worse.

## References

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